

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Computer Science & Engineering

Branch: CSE (AI/CS/DS/CPS)

Semester: III

Mid-Term Examination September-2026

Course: B.Tech. CSE (AI/CS/DS/CPS)

Subject Code: CSE (AI/CS/DS/CPS) - 3003

Max. Marks: 20

Subject Name: Computer Organization and Architecture

Time: 90 Minutes

NOTE:

1. All the questions are compulsory.
2. While solving write question number and its sub part clearly.
3. Solve questions in given sequence and all the sub parts should be at same place.

1.A	Identify whether the following register transfer statements is correct or incorrect? If incorrect then explain what is wrong with it? i. $xT: AR \leftarrow AR', AR \leftarrow 0$ ii. $yT: R1 \leftarrow R2, R1 \leftarrow R3$ iii. $zT: PC \leftarrow AR, PC \leftarrow PC + 1$	03
B.	Show the block diagram of the hardware that implements the following register transfer statement: $yT2: R2 \leftarrow R1, R1 \leftarrow R2$	02
2.A	A computer uses a memory unit with 256K words of 32 bits each. A binary instruction code is stored in one word of memory. The instruction has four parts: an indirect bit, an operation code, a register code part to specify one of 60 registers, and an address part. i. How many bits are there in the operation code, the register code part, and the address part? ii. Draw the instruction word format and indicate the number of bits in each part. iii. How many bits are there in the data and address inputs of the memory?	03
B.	Consider following 16-bit instruction, give the equivalent hexadecimal code and explain the operation performed by the instruction. i. 0001 0000 0010 0100 ii. 1011 0001 0010 0100	02
3.A	Define the following bitwise operations and illustrate each with a suitable 8-bit example. Also explain how the mask is determined in a mask operation. i. Selective set ii. Insert operation iii. Mask operation	03
B.	What is the difference between a micro-processor and a micro program? Is it possible to design a microprocessor without a micro program? Why RTL is preferred for describing internal organization of digital computers.	02

Name of Student: _____

Scholar No.: _____

4.A. ✓	Imagine that you have installed a computer game on your system and want to play it. Based on the memory hierarchy and storage organization of a modern computer system, answer the following questions: i. In which type of memory reserved bootable information resides and why? ii. Where is the computer game stored before you start playing it? Why? iii. When you launch the computer game, where is it loaded for execution? Why? iv. Why is SSD/Hard disk preferred over RAM for secondary storage? Justify it.	03
B.	An 8-bit register contains the binary value 10011100. What is the register value after an arithmetic shift right? Starting from the initial number 10011100, determine the register value after an arithmetic shift left, and state whether there is an overflow.	02